

STRATEGY REPORT · OCTOBER 2026

AI Innovation & Venture Blueprint

From engineering projects to a market-ready startup

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LOCATION	Cairo, Egypt
EVIDENCE DATE	Sources accessed 3 October 2026

YOUSEF	YOUSEF	YOUSEF	MAZEN	TEAM
Guardian RAG	AutoFT	KRONOS	Swarminal	Nest Invest
Permission-aware retrieval · working	Drift-gated fine-tuning · experimental	Agent platform · source private	CLI-agent desktop · owner's project	Graduation project · pending

A discussion document, not an offer of securities or legal advice. Distinguishes verified evidence, user-provided information, analysis and unvalidated hypotheses throughout.

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How to read this report

Every material statement carries one of four labels, so a reader can tell evidence from opinion at a glance:

Label	Meaning
VERIFIED	Checked against a public repository, README, portfolio site or official page on 3 October 2026. "Verified" means the source <i>states</i> it; source code was not executed for this report.
USER-PROVIDED	Stated in the brief that commissioned this report and not independently confirmed online.
ANALYSIS	The authors' assessment, with its basis given beside it.
HYPOTHESIS	A commercial or market assumption that has not been validated with customers and should be tested before it is relied on.

Numbered superscripts such as ^[3] link to the source list in the Appendix, which records each URL and its access date. Vendor-published comparisons are marked as such in the Appendix because they carry a commercial bias.

EXECUTIVE SUMMARY

A strong technical portfolio that has not yet met a customer

The work inspected for this report is unusually well documented for personal R&D: real measurements, stated limitations, and CI-gated security tests. What is missing is any evidence of a buyer. The next 90 days should therefore convert engineering credibility into customer evidence, not into more platform.

5

projects reviewed; 3 owned by Yousef with public evidence, 1 owned by Mazen, 1 with private code

0

paying customers, pilots, letters of intent or funding found or claimed in any source

1

project recommended to lead: Guardian RAG, as a service-backed proof-of-concept

Key findings

- 1 **Guardian RAG is the clearest first offer.** It solves a problem buyers already recognise — staff need policy answers, but restricted content must not leak — and it already enforces access inside retrieval, backed by PostgreSQL Row-Level Security and a hash-chained audit log **VERIFIED** [3]. It is demonstrable in ten minutes with sample documents. The market is crowded, so it should be sold first as a *self-hosted implementation service*, not as a SaaS product **ANALYSIS**.
- 2 **AutoFT makes LLM agents self-maintaining.** It closes the loop that drift monitors and fine-tuning tools each leave open: detect drift in real traffic, decide whether retraining is worth the GPU time, retrain with QLoRA locally, and *ship only if the new model measurably beats the incumbent* **VERIFIED** [4]. Its first real run improved relevance by +0.0486 and was correctly held back as within noise. The next step is better measurement and a first team with a live agent to run it on — today it is a strong basis for evaluation and "should we retrain?" engagements **ANALYSIS**.
- 3 **KRONOS is impressive and too broad.** Seven completed phases, 96 offline tests and nine ADRs are reported, but the source is private, so readiness cannot be independently verified **VERIFIED AS CLAIMED** [5]. As a general agent platform it competes with mature open-source builders; it should be narrowed to one workflow or used as Yousef's delivery toolkit **ANALYSIS**.
- 4 **Swarminal belongs to Mazen and needs its own validation.** It is an Electron/node-pty desktop app for running CLI coding agents side by side on Windows **VERIFIED** [6]. Its README declares MIT, but no LICENSE file could be retrieved, and its screenshots carry a different product name ("BridgeMind-PTY"). Any joint use requires Mazen's written agreement first.
- 5 **Nest Invest cannot be assessed.** The code is private; public sources confirm only an LSTM forecasting service inside a MERN simulator, graded A, and that Osama Ashur built its front end **VERIFIED** [1][10]. Treat it as a delivery credential.
- 6 **Intellectual-property hygiene is the most urgent non-technical risk.** Yousef's public profile lists RAG security architecture and an autonomous fine-tuning system among his R&D tracks at his former employer, YOXEL Technology — the same themes as Guardian RAG and AutoFT [1]. Osama currently works at YOXEL [10]. Independence of the personal repositories, and any employment restrictions, should be confirmed with an Egyptian lawyer before anything is commercialised.

RECOMMENDED SEQUENCE

Days 1–30: align the three people, inventory IP, publish one Guardian RAG demo with synthetic documents, run 12–15 problem interviews. **Days 31–60:** one scoped, paid or co-funded Guardian PoC; offer AutoFT-style evaluation audits as a secondary service. **Days 61–90:** decide — productize, continue consulting, apply to an accelerator, or pause — against the evidence gates on page 33.

METHOD

Evidence base, method and limitations

All supplied links were opened on 3 October 2026. Repositories were inspected through their public pages and READMEs; source code was read where the README reproduced it, but no project was cloned, built or executed for this report.

Source supplied or found	Status	What it contributed
Yousef — GitHub profile README [1]	ACCESSED	Experience, stack, project summaries, statement that most work lives in private repositories.
Yousef — portfolio site [2]	ACCESSED	Consistent with GitHub; no additional project metrics.
guardian-rag repository [3]	ACCESSED	Architecture, ACL model, verified role matrix, roadmap, MIT licence. 3 commits.
AutoFT repository [4]	ACCESSED	Full results, training stats, limitations, dashboard screenshots, MIT licence. 20 commits.
kronos-showcase repository [5]	ACCESSED	Case study only; implementation private. Screenshots, ADR list, test counts, Apache-2.0.
Nest Invest source	PRIVATE	No repository supplied; code kept private by the owner [1] .
Swarminal repository [6]	ACCESSED	README, architecture, screenshots, 8 commits, Windows binary in repo.
Swarminal LICENSE file [7]	404	README badge says MIT; the linked file returned "not found".
Mazen — GitHub profile [8]	ACCESSED	Name confirmed; 19 repositories; no bio, location or links.
Osama — GitHub and portfolio [9] [10]	ACCESSED	Roles, stack, six portfolio projects, Nest Invest front-end credit.
Osama — LinkedIn [11]	BLOCKED	Not retrievable (robots.txt). Nothing from LinkedIn is used.
Yousef — LinkedIn	NOT OPENED	Linked from GitHub but not supplied; same access restriction expected.

Limitations that bound this report

- **No market validation exists in any source.** Every statement about customers, willingness to pay or segment fit is a hypothesis, labelled as such.
- **Self-reported metrics.** Test counts, training results and the Guardian role matrix come from the projects' own READMEs. They are specific and internally consistent, but were not reproduced here.
- **Competitor information** comes partly from vendor-published comparisons (Onyx, FutureAGI, AgentsRoom). These are flagged in the Appendix and should be re-checked before being quoted to investors.
- **No legal advice.** Part 7 identifies questions for Egyptian legal and tax counsel; it does not answer them.
- **No financial model.** Part 8 gives cost categories and illustrative arithmetic from official pricing pages, not a forecast.

OBSERVATIONS FOUND DURING INSPECTION — FIX BEFORE SHARING DEMOS

- AutoFT's published drift-dashboard screenshot shows an unhandled KeyError in the "Query Distribution Comparison" panel [\[4\]](#).
- Swarminal's screenshots are titled "BridgeMind-PTY" and show a "Codex Pro — Upgrade" panel and personal email addresses inside terminal panes (blurred in this report) [\[6\]](#).
- Guardian RAG's public history is 3 commits — normal for a cleaned export, but reviewers may read it as "new". A short development log would help.

PART 01

People

Who is involved, what the public record shows each person has done, and what the group is missing. Nobody named here has agreed to form a company, committed time, or accepted a founding role.

1.1 Founder profile — Yousef Mamdouh

AI/ML engineer, Cairo. B.Sc. Computer Science (Data Science major), Canadian International College (CIC), 2020–2024 VERIFIED ^{[1][2]}. Currently Data Science & AI Team Lead at GDG on Campus CIC USER-PROVIDED.

Role (public profile)	What it evidences
AI/ML Engineer — R&D Execution & Team Lead , YOUXEL Technology, Oct 2025 – Jun 2026	Led multi-track R&D on an enterprise agent platform (TORUK): RAG security, autonomous fine-tuning, self-service fine-tuning POC, agent quality scoring, sports computer vision; supervised a junior engineer; Jira Scrum delivery.
Business Data Analyst , GIG Egypt Life Takaful, Mar – Oct 2025	Jira Service Management, ScriptRunner automation, BI dashboards — exposure to a regulated enterprise environment.
Backend intern , GIG Egypt, 2023	Laravel/MySQL REST API.

All roles and dates as stated on GitHub and the portfolio site ^{[1][2]}; not checked against employer records.

Technical strengths ANALYSIS

- **Retrieval with access control** — designed into the query, not bolted on.
- **Evaluation discipline** — held-out splits, paired comparisons, refusing to ship marginal gains.
- **Local-first LLM systems** — Ollama, QLoRA on an 8 GB GPU.
- **Documentation** — ADRs, runbooks, stated limitations.

Gaps to complement ANALYSIS

- Selling, pricing and contracting.
- Customer discovery in a specific industry.
- Operations: finance, legal, hiring.
- Production operations at scale (on-call, SLAs) — not evidenced in public sources.

Potential contribution as a technical founder ANALYSIS

Yousef's portfolio supports a credible **CTO-type** role for an applied-AI venture: setting architecture, owning evaluation, and translating vague "add AI" requests into measurable systems. The strongest differentiator is not any single project but a *habit* visible across all of them — measuring before trusting. That habit is what enterprise buyers of RAG and fine-tuning are short of. What the record does not yet show is commercial ownership: no client delivery under his own name, no pricing decisions, no sales cycle. A business-side co-founder or advisor would close the most important gap.

1.2 Osama Ashur

Software Engineer, YOUXEL Technology (Jan 2026 – present), previously Frontend Engineer there (Sep 2025 – Jan 2026). B.Sc. Computer Science, Cloud & Mobile Computing, CIC, 2020–2024 VERIFIED ^[10].

Portfolio evidence: React + Supabase hotel management app; TypeScript HR attendance system (React, Vite, Express, MongoDB); Express course API with role-gated routes; a TypeScript full-stack media platform; a .NET

Verified skill	Potential responsibility (to confirm)
React, Next.js, TypeScript	Product UIs for demos and client work
Node/Express REST, JWT, role-gated routes	API layer and auth for client deliverables
MongoDB, MySQL, Supabase	App data models outside the AI core

Core Web API [\[9\]\[10\]](#). Built the **Nest Invest front end** (React, Tailwind, Recharts, React Query).

Verified skill	Potential responsibility (to confirm)
Performance, code splitting, a11y	Polishing demos to client-ready quality

CONSTRAINT

Osama is in full-time employment. His time commitment and any employer restrictions on outside work are unknown and must be clarified before he is named in a proposal.

1.3 Mazen Sarwat Ismail

GitHub account sspotter, 19 public repositories; pinned work includes an Arabic sentiment-analysis dashboard, an AI live-tracker and an agent expense tracker, all TypeScript **VERIFIED** [\[8\]](#). No bio, employer or location is listed. He is the **owner of Swarminal** (Part 2.5). Evidence of desktop engineering: an Electron + React + node-pty application with a documented main/renderer architecture [\[6\]](#). Any role in a joint venture is **subject to mutual agreement** and is not assumed here.

1.4 Team capability map

CAPABILITY	Yousef	Osama	Mazen	Gap?
RAG, retrieval and access control				Covered
LLM fine-tuning and evaluation				Single point
Agent orchestration (server side)				Covered
Python backend / FastAPI				Covered
React / TypeScript product UI				Strong
Node / Express APIs and auth				Covered
Desktop apps / terminals (Electron, PTY)				Single point
Cloud ops, security hardening, SLAs				Gap
Sales, pricing, contracts				Critical gap
Industry domain access (HR, finance, legal)				Critical gap

evidenced by public work listed skill or partial evidence no public evidence (may still exist)

Figure 1. Capability coverage, based only on public repositories and portfolios inspected on 3 Oct 2026. Yousef's domain-access mark reflects enterprise work at previous employers; Mazen's orchestration mark reflects agent-launch logic in Swarminal.

Proposed structure — for discussion, not agreed **PROPOSAL**

Person	Proposed focus	Conditions before it applies
Yousef	Technical lead: architecture, AI systems, evaluation, client technical discovery.	Confirms availability alongside job search and GDG role; confirms IP independence of Guardian RAG and AutoFT.
Osama	Product engineering: UI, APIs, integration, demo polish.	Agreed hours per week; employer's outside-work policy checked.
Mazen	Developer-tooling and desktop track; owner of Swarminal.	Separate decision on Swarminal; written agreement before any shared use.
To recruit	Business lead or advisor: discovery, pricing, sales, contracts.	Ideally someone with buyer access in one target industry.

Assessment **ANALYSIS** : AI, web and desktop are covered with little overlap — good for delivery, fragile for resilience. The critical gap is commercial: nobody has publicly evidenced selling, pricing or contracting.

PART 02

Project portfolio and market analysis

Five projects, two owners, very different levels of evidence. Each is assessed on the same 17 points, then compared on transparent criteria.

2.1 Portfolio at a glance

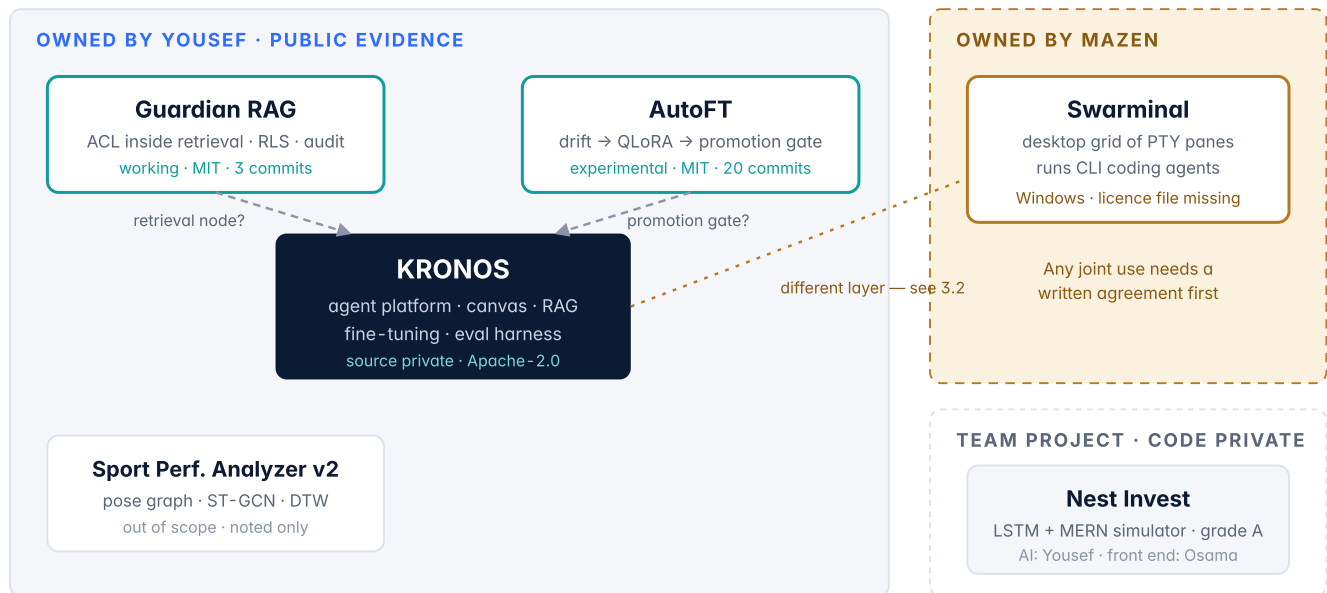


Figure 2. Project ecosystem. Dashed arrows are *possible* integrations discussed in Part 3, not existing ones. Sport Performance Analyzer appears on Yousef's profile but was not in the brief and is not analysed.

Project	Owner	Evidence available	Stated maturity	One-line verdict (analysis)
Guardian RAG	Yousef	Full public source, CI, screenshots	Core features working; PDF/DOCX, hybrid search, SSO planned	Lead with it — clearest buyer pain, easiest demo.
AutoFT	Yousef	Full public source, real run results	"Experimental"; promotion gate returned HOLD	Self-maintaining agents; prove on a client agent first.
KRONOS	Yousef	Case study, screenshots, 3 code excerpts	Phases 0–6 "Done" (self-reported)	Use as delivery toolkit; narrow before selling.
Swarminal	Mazen	Public source, binary, screenshots	Release labelled v77.77.77; 8 commits	Mazen's call; validate with Windows developers.
Nest Invest	Team (incl. Yousef, Osama)	Profile text only	Graduation project, grade A	Portfolio credential; analysis pending.

2.2 • OWNER: YOUSEF MAMDOUH • MIT LICENCE

Guardian RAG — access control that runs inside retrieval

A self-hosted assistant that answers questions over company policy documents using only the passages the asker is allowed to read, cites them, and records every answer and denial in a tamper-evident log.

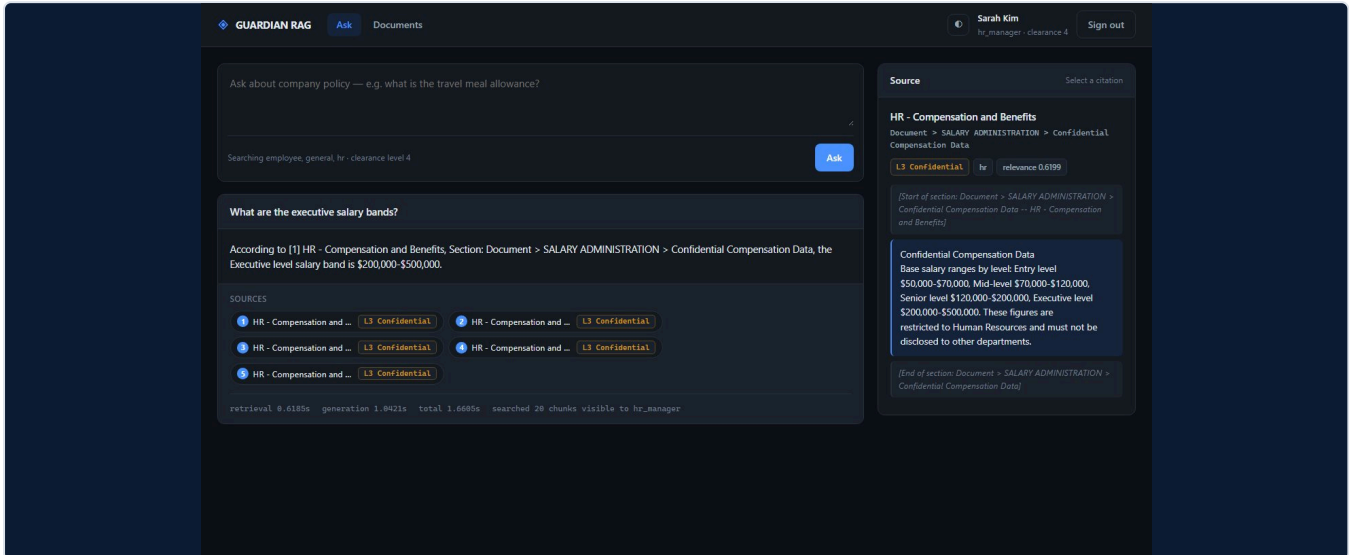
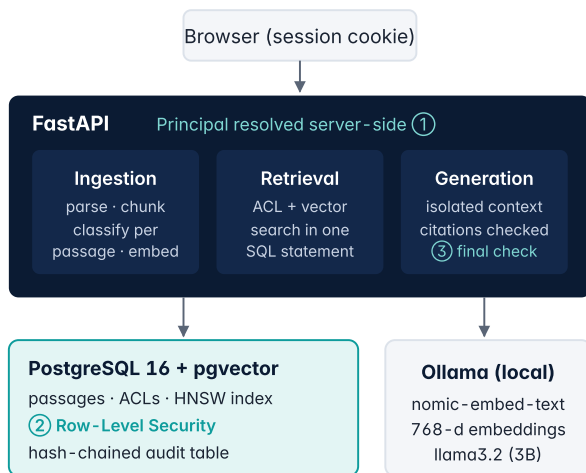


Figure 3. Real screenshot from the repository: an `hr_manager` asks for executive salary bands and receives a cited answer from an L3-Confidential passage. Source: docs/images/05-hr-authorized.png^[3]. The same question is withheld for `finance_manager` and `employee` roles.

How it works VERIFIED



Redrawn from the README architecture block^[3]. ①②③ are the three enforcement layers the README describes. A passage is returned only if domain grant, clearance level and role allow-list all pass.

Implementation status VERIFIED

Working (per README)	Planned
Auth and sessions; upload with content-hash idempotency and versioning; section-aware chunking; batched embedding; ACL-filtered retrieval; cited answers; hash-chained audit; admin console; CI-blocking security tests over the full role × document matrix.	PDF/DOCX ingestion with OCR; background ingestion worker; lexical + vector fusion with re-ranking; streaming answers; SSO.

Tested scale: 3 policy documents, 29 passages, 6 roles^[3]. This proves the access model, not retrieval quality or performance at enterprise volume.

WHY IT IS THE LEAD CANDIDATE ANALYSIS

The failure it prevents — a plausible question returning salary bands — is concrete, explainable to a non-technical HR or compliance buyer in one sentence, and demonstrable live by switching roles. Few demos make a risk this visible.

2.2 · GUARDIAN RAG — COMMERCIAL PROFILE

Problem addressed	Organisations need staff to self-serve on policy (leave, expenses, benefits) while keeping sections such as compensation restricted. Post-retrieval filtering lets restricted text influence ranking; file-level classification hides benign content VERIFIED AS STATED [3].
Target customer	HYPOTHESIS Mid-sized organisations (≈200–2,000 staff) in confidentiality-heavy sectors — private healthcare groups, law firms, private universities, fintech — that have internal IT but no AI team, and are uneasy sending documents to third-party APIs.
Customer pain points	HYPOTHESIS HR and finance teams answering the same questions repeatedly; policies that exist but are not read; compliance reluctance to adopt cloud AI assistants; no audit trail for who saw what.
Competitors & alternatives	Enterprise search/assistants (Glean, Cohere North), open-source platforms (Onyx, MIT, retrieval-time permissions), cloud building blocks (Bedrock Knowledge Bases, Azure AI Search), frameworks (LlamaIndex, LangChain) [12]; manual: intranet search, emailing HR. See 2.7.
Differentiation hypotheses	HYPOTHESIS (1) per-passage classification so one handbook serves several audiences; (2) enforcement in the database as well as the app; (3) silent denial that does not confirm restricted content exists; (4) fully local operation; (5) local delivery and Arabic-region support. Note that (1)–(4) are features, not moats: Onyx also claims retrieval-time permissions [12]. Durable advantage would have to come from delivery, domain templates and trust.
Revenue models	(a) Fixed-scope PoC; (b) implementation project; (c) monthly support and content-update retainer; (d) later, licensed on-prem package. SaaS is the weakest fit for a "data never leaves" proposition.
MVP scope	Existing core + PDF/DOCX ingestion + SSO or directory sync with one identity provider + Arabic-language retrieval test + deployment script for a customer server. Hybrid search can wait unless the pilot data demands it.
Customer PoC plan	2–4 weeks; 10–30 of the client's non-sensitive policy documents; 3–5 roles; 50 agreed test questions incl. "should be refused" cases; success = zero leaks in the refusal set, ≥ agreed answer-accuracy threshold judged by the client's HR owner, deployment on client infrastructure.
Effort & dependencies	ESTIMATE MVP additions ≈ 4–8 person-weeks. Depends on: client server or VM (GPU optional for a 3B model, CPU latency to be measured), identity provider access, Arabic document quality, and the IP clearance in Part 7.
Main risks	Crowded category; buyers may prefer the assistant bundled in their existing suite; small local models may answer poorly in Arabic; a single leak would be fatal to trust; long enterprise sales cycles.
Validation experiments	12+ interviews with HR/compliance/IT heads; ask each to name a document they would <i>not</i> put in a cloud assistant; offer the demo and track who requests a PoC; run the public demo with a red-team prompt challenge.
Continue / pivot / pause	Continue if ≥ 3 organisations request a PoC with their own documents within 60 days. Pivot to a generic RAG implementation service if interest is in "chat with our docs" but not access control. Pause productisation if no one will share documents even under NDA.

RECOMMENDED FIRST SEGMENT **HYPOTHESIS TO TEST**

Private healthcare groups and private universities in Egypt, starting with HR, finance and internal-policy documents. Reasoning: both hold documents with clear audiences (staff vs management; students vs faculty vs administration), both are wary of sending them to cloud AI tools, and Yousef's CIC and GDG on Campus connections may give warm introductions on the university side. None of this proves demand — it only makes the first 15 interviews cheaper to get. Law firms are the alternative if neither responds.

2.3 · OWNER: YOUSEF MAMDOUH · MIT LICENCE

AutoFT — self-maintaining LLM agents

Automatic fine-tuning with drift detection and evidence-based promotion, for LLM agents running on local hardware. AutoFT is a closed loop around a deployed agent: it watches the questions people actually ask, detects statistically when that distribution has shifted, decides whether the shift justifies the cost of retraining, runs a real QLoRA fine-tune — and **refuses to ship the result unless it measurably beats the model already in production** on held-out data^[4].

Everything runs on one machine: Ollama serves the model and a single consumer GPU trains it — no API keys, no cloud, no data leaving the host. The worked example is a customer-support agent built on 5,000 real public question-answer pairs; the pipeline itself is domain-agnostic, and `scripts/ingest_documents.py` builds a training set from any folder of documents **VERIFIED**^[4].

The problem: an agent that quietly gets worse

An agent works well on launch day. Six months later it is worse and nobody can say when that happened — customers ask about new products, a regulation changes the correct answer, the customer base drifts into different questions. The model has not changed; the world around it has. Three things go wrong, in order:

1 · Nobody notices

Quality decays gradually. There is no alarm — only a slow rise in complaints.

2 · Nobody knows when to act

"Retrain it" is a guess. Retraining costs GPU hours and risks shipping something worse.

3 · Nobody proves it helped

A new model ships because training finished, not because it was better — and the old one is already gone.

Existing tools each solve a third of this: fine-tuning platforms train models but do not say *when*; drift monitors raise alerts but do not *act*. Neither closes the loop, and neither proves the retrain was worth the money. AutoFT is built to close it.

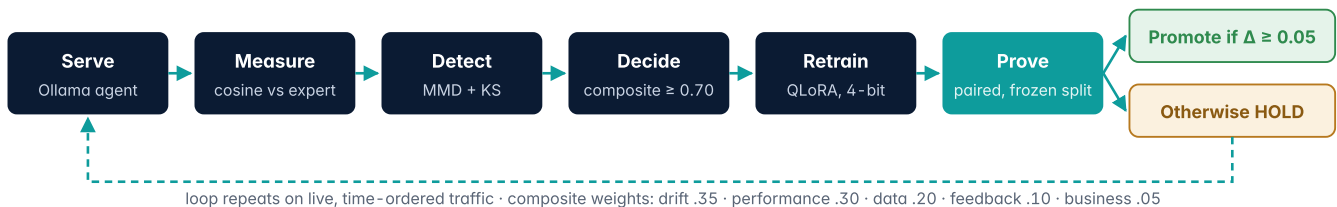


Figure 4. The AutoFT loop — serve → measure → detect drift → decide → retrain → prove it helped → promote or reject — redrawn from the README stage table^[4].

You have	AutoFT gives you	Stage	What happens
An LLM agent in production that "feels worse than it used to"	A statistical answer to <i>when</i> it changed, and by how much	Measure	Each answer scored against a reference in embedding space
Pressure to retrain, and no budget to do it blindly	A composite score that says whether the evidence justifies GPU time	Detect	MMD + KS tests over a rolling window of real, time-ordered traffic
A fine-tuned model and no idea if it is actually better	A paired head-to-head against the incumbent on held-out data	Decide	Weighted composite of drift, performance, data volume, feedback, business priority
Sensitive data that cannot leave your network	A loop that runs entirely on local hardware	Retrain	QLoRA — 4-bit frozen base, trainable adapters
		Prove	Head-to-head vs incumbent on a frozen, hash-guarded test split
		Promote	Ships only if it clears the threshold. Otherwise: HOLD

THE POINT

A model that does not measurably improve does not ship. That single rule is what separates AutoFT from a retraining script.

2.3 • AUTOFT – EVIDENCE FROM THE FIRST REAL RUN



Figure 5. Real dashboard capture (cropped) from the repository: loss curves, run settings and the promotion check returning HOLD at +0.0486 [4].

The one real run VERIFIED AS REPORTED

Base → llama3.2:3b → LoRA-tuned adapter v1
tuned

Data	3,998 training pairs from a public customer-support Q&A dataset; 501-question frozen test split, 50 sampled
Mean relevance	0.7736 → 0.8221 (+0.0486)
Win rate	54% (27 W / 23 L)
Gate	HOLD — needed +0.0500
Compute	2 h 04 m, RTX 5060 8 GB, peak 6.0 GB
Noise	Base scored 0.7736 / 0.7946 / 0.8084 on three identical runs (spread 0.035)

The fine-tune improved the model and was *not* promoted — the system working as designed. The README calls the gain real but small, inside measurement noise; the stated fix is greedy decoding over all 501 questions, not a lower threshold.

WHAT THIS EVIDENCE DOES AND DOES NOT SUPPORT

It supports: a working end-to-end loop; correct statistical drift tests validated on a known shift; a gate that rejects marginal candidates. It does **not** yet support claims that automated retraining saves money or improves quality in production — the only run did not clear the gate, user feedback is not collected, and cosine similarity is acknowledged as a weak proxy [4].

Commercial profile

Technical architecture	FastAPI service, Streamlit dashboard, SQLite/TinyDB storage, Ollama serving, separate CUDA training environment (PEFT, bitsandbytes, transformers), llama.cpp GGUF conversion; MMD (RBF, median heuristic) and per-component KS over a 50-d PCA projection [4].
How it differs	Drift monitors (Evidently, Arize Phoenix, NannyML, WhyLabs) detect and alert; a vendor comparison lists none of them as retraining automatically outside enterprise tiers [17]. Fine-tuning tools train but do not decide <i>when</i> or prove <i>whether</i> . AutoFT's distinctive element is the evidence gate that can say no .
Integration path	Wraps any Ollama-served model; would need connectors for MLflow-style registries, OpenTelemetry traces, and non-Ollama serving (vLLM, cloud endpoints) to fit existing ML infrastructure.
Target customer	HYPOTHESIS Teams with an LLM agent in production on private data — support desks, internal help, service operations — that cannot send data out and have, or can rent, a GPU (the "You have" column above).
Business problem	The three failures: decay nobody notices, retraining by guesswork, and shipping without proof [4].
Value proposition	A self-maintaining agent: drift caught statistically, retraining only when the evidence justifies the GPU time, and only proven improvements shipped.

Engineering services	LLM evaluation audit; held-out test-set design; drift monitoring set-up; one supervised fine-tuning experiment with a go/no-go report.
Product direction	HYPOTHESIS A small open-source "promotion gate" library (paired evaluation, frozen hashed splits, noise estimate) usable in CI for any model or prompt change — narrower and more adoptable than the full loop.
MVP requirements	Fix the evaluation noise (temperature 0, all 501 items); add an LLM-judge rubric with calibration; add a feedback UI; run drift checks off the request path; document a second domain.
Demo scenarios	(1) Inject a topic shift and watch live MMD cross the threshold; (2) replay the HOLD decision and explain why it is correct; (3) show the chat-template bug story as an evaluation lesson.
Validation metrics	Gate decision stability across reruns; false-alarm rate of drift detection on unchanged traffic; time from drift to decision; client-judged answer quality, not only cosine.
Effort & dependencies	ESTIMATE 3–6 person-weeks for MVP fixes. Depends on GPU access; Llama 3.2 Community License terms and dataset licences for any commercial use ^[4] .
Risks & limitations	Single-GPU, single-tenant; cosine proxy; few clients have enough traffic for statistical drift; fine-tuning teaches style more than facts (README's own caveat); buyers may simply switch to a newer base model instead of retraining.
Validation experiments	Ask 10 teams running LLM features how they decide to change models today; offer a fixed-price evaluation audit; count requests.
Continue / pivot / pause	Continue as service if ≥ 2 audit engagements or serious requests in 90 days. Pivot to the open-source gate library if developers star/use it but nobody pays. Pause productisation otherwise; keep as portfolio.

2.4 · OWNER: YOUSEF MAMDOUH · APACHE-2.0 · SOURCE PRIVATE

KRONOS — a local-first agent platform

A full-stack platform for building, running, evaluating and fine-tuning multi-agent systems: a React Flow canvas, a DAG workflow engine, hybrid RAG, a six-provider LLM adapter, a fine-tuning panel and a code-review agent. The public repository is a case study; the implementation is private and available for review on request ^[5].

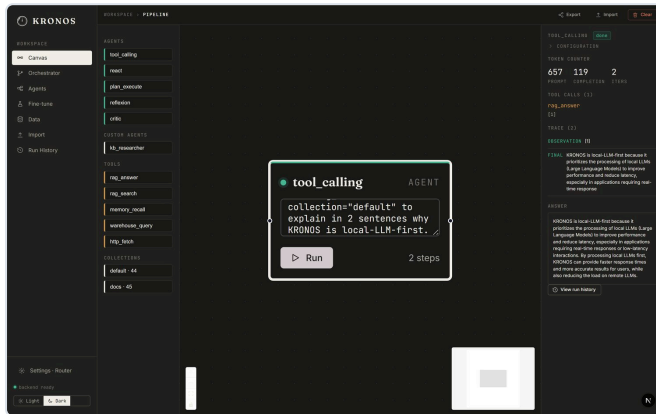


Figure 7. Canvas: a `tool_calling` agent invoking `rag_answer`, with tokens and trace in the inspector. Real screenshot ^[5].

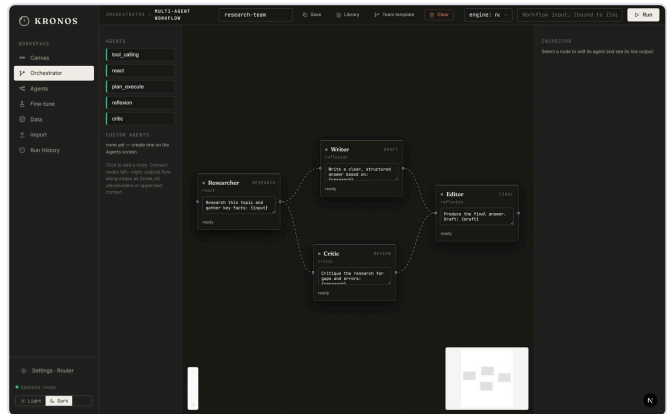


Figure 8. Orchestrator: a four-node "research-team" DAG (react → critic → reflexion writer → editor). Real screenshot ^[5].

Reported as done (self-reported, not verifiable)	Evidence offered	Not evidenced / open
Phases 0–6: LLM adapter, 12 source types, hybrid RAG with RRF, MCP tools and 5 agent strategies, canvas with Redis Streams, LoRA/DPO training, project import	Screenshots of a live instance; 3 verbatim code files; raw mypy/ruff/pytest output in docs/evidence; 9 ADRs; 96 offline tests	Multi-user auth and tenancy; deployment beyond Docker Compose on one machine; performance under load; any external user

The breadth problem ANALYSIS

KRONOS covers ground that whole companies split between them: workflow builders (Dify, Langflow — each reported at well over 100k GitHub stars ^[15]), observability tools, fine-tuning services and code-review agents. As a general platform it would need years of hardening to compete. A useful market signal: Flowise, another open-source LLM builder, archived its repository on 13 August 2026 ^[16] — visual-builder vendors find this category hard to monetise even with large communities. KRONOS's value today is as **proof of engineering depth** and as **Yousef's own delivery toolkit**.

Target customer	HYPOTHESIS Not an end-buyer product yet. Near-term "customers" are evaluators of the team: technical due-diligence reviewers, CTOs hiring an AI partner.
Value proposition	Shows the team can build and operate the whole stack locally, without SDK lock-in.
Competitors & alternatives	Dify, Langflow (open source); n8n-style automation tools; LangGraph-based custom builds; Flowise (archived) ^{[15][16]} .
Differentiation hypotheses	Local-first by default; evaluation and fine-tuning in the same loop as orchestration; run history exported directly to training data.
Revenue models	Indirect: client PoCs delivered on KRONOS; source licence or support for a client that adopts it. Direct SaaS not recommended now.
Focused first use case & MVP	One packaged workflow: "policy question → Guardian-style permission-aware retrieval → cited answer → deterministic eval gate", shipped as a template a client can inspect on the canvas. Only build it after Guardian has a pilot asking for multi-step tasks (e.g. drafting a reply that cites policy).

Public demo	Recorded 3 -minute canvas walkthrough plus a read -only hosted canvas if feasible; live only in meetings.
PoC plan	Only as the engine behind a Guardian or agent -development engagement, never sold as "the platform".
Effort & dependencies	ESTIMATE Multi-tenant hardening is a multi-month effort; the template workflow is ≈ 2–3 person - weeks once Guardian retrieval is wrapped as an MCP tool. Depends on Redis, Qdrant, Postgres, Ollama.
Risks	Scope creep; maintenance load of six providers and 12 connectors; private source limits trust; the IP question in Part 7.
Validation experiments	Offer source review to two CTOs and ask what they would pay to have built on it; track whether demos generate requests for <i>the platform</i> or for <i>a specific workflow</i> .
Continue / pivot / pause	Freeze new platform features for 90 days. Resume only for a feature a paying engagement needs.

2.5 · OWNER: MAZEN SARWAT ISMAIL · LICENCE UNCLear

Swarminal — a desktop cockpit for CLI coding agents

OWNERSHIP

Swarminal is **Mazen's existing project**. Nothing in this report implies joint ownership, and no assumption is made that it can be combined with the other projects. Any shared development, licensing or commercial use requires Mazen's written agreement (Part 7).

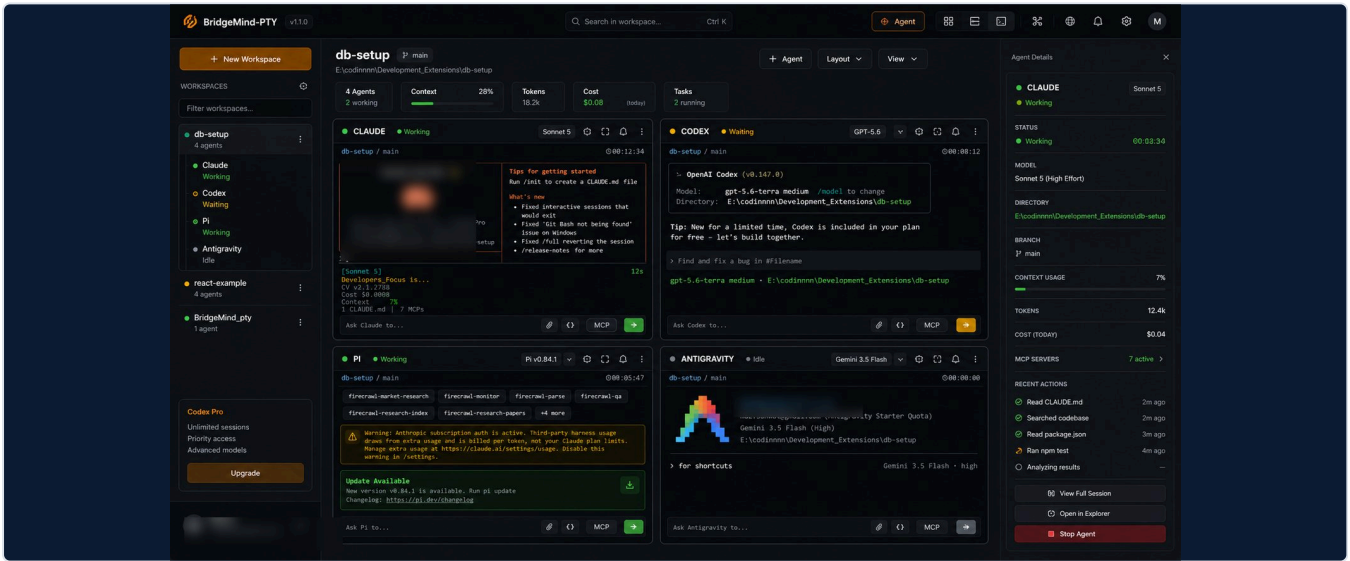


Figure 9. Real screenshot from the repository (he-ro -app .png): four panes running Claude Code, Codex, Pi and Antigravity in one workspace, with an agent-details panel. Personal identifiers blurred for this report. Note the window title reads "BridgeMind-PTY v1.1.0"^[6].

Architecture VERIFIED AS DOCUMENTED

- **Electron main process (Node.js):** PtyManager on node-pty/ConPTY with a 64 KB replay ring; AutoLaunchTracker with an agent allow -list; background Git status polling; debounced atomic JSON session store; WebContentsView browser panel.
- **Renderer:** React + TypeScript + Zustand; react-mosaic tiling; xterm.js with WebGL.
- **Agents:** presets for claude, codex, gemini, pi; any CLI via agents .json; opt-in unattended mode with - - dangerously-skip-permissions.

Development status

Commits	8; 1 star; 0 forks
Platform	Windows 10+ (ConPTY)
Distribution	. exe committed to the repo; no GitHub Release
Licence	README badge: MIT · LICENSE file: 404 ^[7]
Version label	"v77.77.77"; screenshots say v1.1.0
Tests	npm test documented; coverage not shown

- **UX:** Ctrl+Shift keyboard layer, broadcast typing, pinned panes, close-guards, localhost port detection.

2.5 · SWARMINAL — ANALYSIS

Problem addressed	Developers who run several CLI coding agents in parallel juggle terminal windows, lose sessions, and miss which agent is waiting for input.
Target users	HYPOTHESIS Windows-based developers and small teams already paying for one or more CLI agents; agencies running many client repos.
Competitors & alternatives	Claude Squad (open-source TUI, macOS/Linux); Conductor (macOS app plus cloud sandboxes); Crystal (MIT Electron app — <i>deprecated Feb 2026</i>); Vibe Kanban; Sculptor; Omnara; Cursor Cloud Agents; AgentsRoom (Mac/Windows/Linux, free tier of 3 projects / 6 agents) ^{[13][14]} . Manual: Windows Terminal split panes, tmux.
Differentiation hypotheses	(1) Windows-native ConPTY — several listed rivals are macOS/Linux only ^[13] ; (2) agent-agnostic via a JSON registry; (3) embedded browser preview with port detection; (4) safety guards against killing busy agents. AgentsRoom already covers Windows, so (1) is narrowing.
Productivity benefit	Plausible but unmeasured. A credible claim needs a timed task study (e.g. same three-repo change with and without Swarminal).
Business models	Open-source tool (sponsorship, reputation) — most realistic start. Premium app (cost and token dashboards, session sync) — feasible but competes with free tools. Team environment — needs a server component it does not have. Workflow integration — e.g. launching panes from a task board. Enterprise support — premature.
MVP / readiness work	Add a LICENSE file; move binaries to GitHub Releases; sign the installer (Windows SmartScreen); settle the product name; remove personal data from screenshots; publish a 2-minute video.
Main risks	The agent vendors ship their own multi-session UIs; fast-moving CLIs break presets; a close architectural analogue (Crystal) was deprecated; unattended "dangerous mode" creates liability if marketed carelessly.
Validation	Release publicly; target 100 installs and 20 weekly-active users in 60 days; interview 10 users about what they would pay for.
Continue / pivot / pause	Mazen's decision. Continue if weekly-active use grows without paid promotion; pivot to a plugin for an existing terminal if users value only one feature; pause if a major agent vendor ships an equivalent Windows UI.

Relationship to KRONOS ANALYSIS

The two operate at different layers. KRONOS orchestrates *API-level* agents on a server canvas; Swarminal supervises *CLI processes* on a developer's desktop — its own README frames this as "orchestrate real CLI processes, not raw API model calls" ^[6]. A combined product would serve two different buyers at once. The only natural touchpoint is narrow — e.g. a KRONOS workflow node that spawns a Swarminal pane — and it should wait until each has its own users.

2.6 · TEAM GRADUATION PROJECT · SOURCE PRIVATE

Nest Invest — analysis pending

Public sources are not enough to assess architecture, market or commercial potential. Per the brief, this section records what is known, what is missing, and a template to complete.

What is verified

- Graduation project at CIC, grade A **VERIFIED** ^[1]
- Full-stack web app: users trade with a virtual budget on real market data ^[1]
- LSTM price-forecasting model served from a separate Python ML service; Yousef owned model design, training and integration "delivered in one month" ^[1]
- Stack: PyTorch, pandas, scikit-learn, yfinance; MongoDB, Express, React, Node ^[1]
- Osama Ashur built the front end (React, Tailwind, Recharts, React Query) ^[10]

What is missing

- Source code or architecture diagram
- Full team list and each member's contribution
- Forecast evaluation (metrics, baseline, back-test period)
- University rules on ownership of graduation work
- Licensing of market data used (yfinance is unofficial)
- Any user testing

COMMERCIAL CAUTION ANALYSIS

A price-prediction feature in a consumer trading context raises regulatory questions (investment advice), market-data licensing costs, and model-validity risk. None of that should be explored until the evidence below exists. For now, Nest Invest is best presented as **proof that Yousef and Osama have shipped together**.

Project profile template — to complete

Field	Evidence required
Team and IP	All contributors; university policy on ownership; written consent of every co-author before reuse
Architecture	Service diagram; data flow from market feed to forecast to UI
Model performance	Metric (MAE/RMSE/directional accuracy) vs naive baseline on a held-out period
Data rights	Source and terms for price data; cost of a licensed feed
Users	Who tried it; what they did; retention
Target use	Education (e.g. university finance courses) vs retail; regulatory review for either
Demo	Recorded walkthrough; screenshots approved by all contributors

2.7 • COMPARISON

Comparison matrix

Scores are qualitative (1 = weak, 5 = strong) and record the authors' judgement from the evidence in this report. They are **not** probabilities, valuations or market sizes, and are deliberately not summed.

Criterion	Guardian RAG	AutoFT	KRONOS	Swarminal	Nest Invest	Basis
Clarity of target customer	4	3	2	3	n/a	Can a specific buyer be named?
Ease of demonstrating value	5	3	3	4	n/a	Is the benefit visible in under 5 minutes?
Technical readiness (inspected)	4	3	3	2	1	Public code, tests, stated limits. KRONOS capped: private.
Development complexity to MVP (5 = low)	3	2	1	3	n/a	Effort estimates in each profile
Recurring - revenue potential	3	3	3	2	n/a	Retainers vs crowded free alternatives
Validate with a small pilot	5	3	2	3	n/a	2–4 week scoped test possible?
Independence from external infrastructure	3	2	2	4	n/a	GPU, data stack, third-party CLIs
Fit with team's verified skills	5	4	4	3	3	Figure 1. Swarminal: Mazen only.

READING THE MATRIX

Guardian RAG is strongest on the criteria that matter most in the next 90 days — demonstrability, pilot-ability and team fit. Its weak spot is recurring revenue, which a support retainer can partly address. KRONOS scores high on capability but low on everything that makes it sellable soon. Swarminal is the least dependent on infrastructure and easiest to put in users' hands, which suits an open-source-first test — but that test is Mazen's to run.

Competitor landscape

Project	Direct competitors	Adjacent products	Open-source alternatives	Manual approach
Guardian RAG	Glean; Cohere North; Onyx ^[12]	Bedrock Knowledge Bases; Azure AI Search; Vectara ^[12]	Onyx (MIT); LlamaIndex, LangChain frameworks ^[12]	Intranet search; asking HR; restricted shared drives
AutoFT	None found doing drift → retrain → gate end-to-end in open source	Arize, Fiddler, Datadog LLM monitoring ^[17] ; managed fine-tuning services	Evidently, NannyML, whylogs, Phoenix ^[17]	Ad-hoc retraining when complaints rise
KRONOS	Dify; Langflow ^[15]	Automation suites; LangGraph custom builds	Dify, Langflow; Flowise (archived 2026) ^[16]	Hand-written scripts per workflow
Swarminal	AgentsRoom; Conductor; Claude Squad ^{[13] [14]}	Cursor Cloud Agents; Omnara ^[13]	Claude Squad (AGPL); Vibe Kanban; Sculptor ^[13]	Terminal split panes; tmux

Landscape compiled from sources dated May–August 2026, several vendor-authored. "None found" means none appeared in the sources reviewed, not that none exist.

PART 03

Productization and commercial pathways

Three paths that can run in sequence rather than compete: services fund and inform products; products, once validated, attract partners and capital.

3.1 Path A — AI/ML engineering services

Services are the fastest route to customer evidence and income, and they suit a team with deep skills but no validated product **ANALYSIS**. Each offer below maps to work the team has already evidenced.

Service	Ideal customer	Problem	Deliverables	Boundaries	Evidence base
Permission-aware knowledge assistant	Mid-size firm, policy-heavy, data-sensitive	Staff can't find policy; compliance blocks cloud AI	Deployed assistant, role model, refusal test set, audit view	≤ 3 document domains, 1 identity source	Guardian RAG
RAG implementation / rescue	Startup with a weak "chat with docs" feature	Wrong or uncited answers	Retrieval eval set, chunking and hybrid search fixes, before/after report	Client's existing stack	KRONOS RAG, Guardian
AI agent development	Ops-heavy SME	Repetitive multi-step tasks	One agent workflow with tools, traces, eval gate	One workflow, read-mostly tools	KRONOS
LLM integration	Product company adding AI	No in-house LLM skills	Provider-agnostic adapter, prompt suite, cost guardrails	Backend + one UI surface	KRONOS adapter; Osama
Evaluation & monitoring audit	Team with an LLM feature in production	"Is it getting worse? Did the change help?"	Held-out set, paired eval harness, drift checks, written findings	Fixed 2–3 weeks	AutoFT
Fine-tuning experiment	Domain team with private data	Generic model style/terminology mismatch	One QLoRA run, go/no-go report, merged model if promoted	One base model, one dataset	AutoFT
Workflow automation	Back-office teams	Manual data shuffling	Automated pipeline with human approval steps	Existing systems only	GIG ScriptRunner work
Product development	Founders needing an MVP	No engineering team	Web app (React/Node) with AI feature	Fixed feature list	Osama portfolio
Developer-tooling integration	Dev teams adopting CLI agents	Agent sprawl	Set-up, conventions, multi-agent workflow	Tooling only	Swarminal — <i>only with Mazen's agreement</i>

Common success criteria

- Agreed test set written *before* building
- Measured before/after on that set
- Deployed on the client's infrastructure, handed over with docs
- A named client owner signs off

Common prerequisites and risks

- NDA and data-processing terms before any document is shared
- Client provides representative, non-production data for PoCs
- Scope creep — fix it with a change-request clause
- Team availability — engagements must fit real hours

How to price without inventing market rates

Build each quote from components, then sanity-check against value:

1. **Effort** = estimated person-days × the team's internal day rate (decide it once, together).
2. **Risk buffer** 15–30% for unknown data quality or access delays.
3. **Infrastructure** passed through at cost (GPU hours, hosting, API tokens).
4. **Support** priced separately as a monthly retainer.
5. **Value check**: compare with the client's cost of the problem (hours saved, risk avoided). If the quote exceeds a small fraction of that value, cut scope rather than price.

Gather 5–10 real quotes from Cairo-based agencies and freelancers before fixing a day rate. The percentages above are planning conventions, not market data.

Sample service proposal structure

1. **Context** — the client's problem in their own words, from discovery
2. **Objective** — one measurable outcome
3. **Scope** — in / out lists; documents, roles, systems
4. **Approach** — phases with a demo at the end of each
5. **Acceptance criteria** — test set and thresholds
6. **Deliverables** — code, deployment, docs, report
7. **Client responsibilities** — data, access, a named owner
8. **Team** — named people and confirmed availability
9. **Timeline** — weeks, with decision points
10. **Commercials** — fee, payment schedule, pass-through costs
11. **Data & IP terms** — who owns what; reuse of pre-existing tools
12. **Assumptions & risks** — stated plainly

Discovery questionnaire

Topic	Questions
Problem	Walk me through the last time someone couldn't find a policy answer. Who answered it? How long did it take? What happens when the answer is wrong?
Current approach	What have you already tried — intranet, shared drives, a chatbot? Why did it stop?
Sensitivity	Which documents would you never put into a cloud AI tool? Who decides that? Is there an audit or regulatory requirement for who accessed what?
Users & roles	How many staff? How many distinct permission levels? Where is identity managed (AD, Google, other)?
Data	Formats (PDF, Word, scanned)? Languages — Arabic, English, mixed? How often do policies change?
Infrastructure	On-premise servers, private cloud or public cloud? Any GPU? Who in IT would own it?
Decision	Who would sponsor a pilot? Who signs? Is there budget this year? What would make a 4-week pilot a success for you personally?

3.2 Path B — product startup, validation first

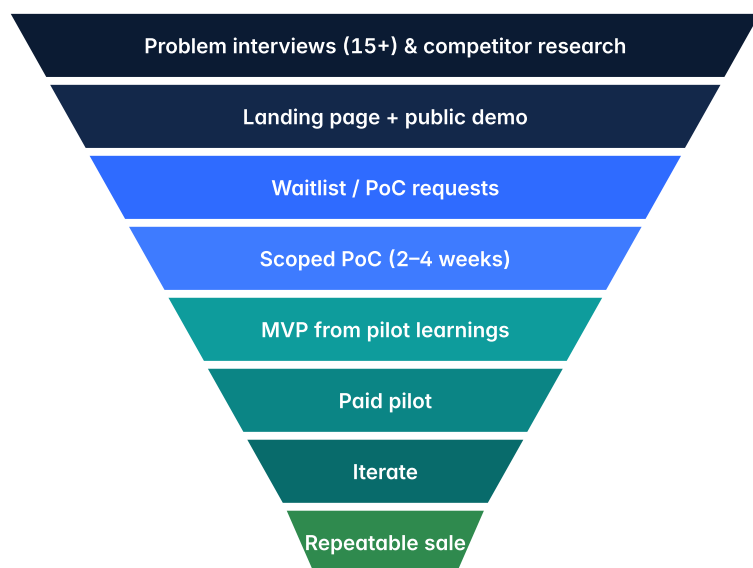


Figure 10. Validation funnel. Each stage has a gate; skipping stages is the most common way technical teams build products nobody buys.

Recommended candidate **HYPOTHESIS**

A **self-hosted, permission-aware policy assistant** for regulated mid-size organisations, built on Guardian RAG. Start as services; productise the repeatable parts (deployment script, role templates, refusal test suites per industry) only after 2–3 deployments.

When to keep projects independent

- Different buyers (HR/compliance vs developers vs ML teams)
- Different owners (Swarminal)
- No client has asked for the combination

When integration could help

- A pilot asks for multi-step tasks over policies → Guardian retrieval as a KRONOS tool
- A client fine-tunes an assistant → AutoFT's promotion gate as a deliverable

Pairing	Assessment (analysis)
KRONOS ↔ Swarminal	Different layers and buyers (server API agents vs desktop CLI processes); different owners. Keep separate. Revisit only if both have users and a concrete joint use case appears — and then under a written agreement.
Guardian RAG ↔ KRONOS	The most natural fit: KRONOS already has RAG tools and MCP; Guardian's ACL-in-retrieval could become a permission-aware retrieval tool. Build only when a client needs agent workflows on top of governed retrieval.
AutoFT ↔ AI infrastructure services	Strong as a <i>method</i> inside evaluation and fine-tuning engagements; KRONOS already has an eval harness and model registry that overlap. Do not ship as a standalone product until the evaluation-noise fix is done.

3.3 Path C — co-founders, partners and investors

Four stages are often blurred together. Being clear about which one the group is in prevents premature conversations

ANALYSIS

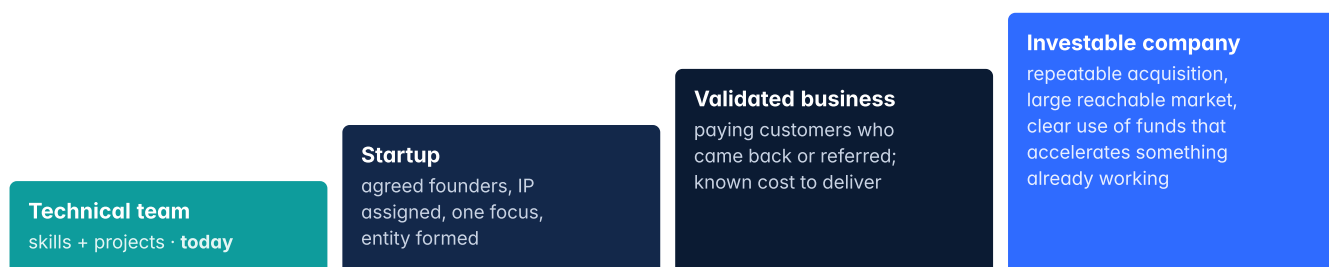


Figure 11. Credibility ladder. The group is on the first step. Each step needs evidence of a different kind, not more engineering.

Audience	What would make the team credible	Evidence that exists today
Business co-founder	Clear technical lead; working demo; honest equity and IP position; a problem with buyers the co-founder knows	Demos and documentation — strong
Mentors	A specific question and 3+ customer conversations to discuss	None yet

Audience	What would make the team credible	Evidence that exists today
Accelerators	Committed founders, an entity or intent to form one, early users or pilots	Engineering only
Angel investors	Paying pilot, founder commitment, sensible use of funds	None yet
Early-stage VC	Repeatable sales motion, retention, market thesis	None yet
Strategic tech partners	Reliable delivery record, security posture, integration with their stack	Security design in Guardian — partial

Before any fundraising conversation

- **Founder alignment:** roles, hours, decision rights, what happens if someone leaves
- **Ownership & IP:** clean assignment of each repo to the entity (Part 7)
- **Market validation:** interview notes and pilot outcomes
- **Technical proof:** live demo, security test results
- **Traction:** at least one paying or co-funded pilot
- **Business model:** priced offer with known delivery cost
- **Funding need & use of funds:** tied to a specific bottleneck (e.g. a sales hire), not "build the platform"

RISKS OF PREMATURE FUNDRAISING

- Valuation set on no evidence makes later rounds harder
- Equity split locked in before roles are proven
- Money spent building features no customer asked for
- Investor reporting distracts from discovery
- Employment and IP issues surface in due diligence

Accelerator applications are lower-risk and useful once founders are committed. ITIDA's Cairo Program for AI startups (with Plug and Play Egypt and NBE) closed its 8th-cohort call on 30 Aug 2026 ^[20] — watch for the next cohort.

PART 04

Demo and deployment strategy

The goal is the fastest *reliable* demonstration of customer value. A recorded demo that always works beats a hosted one that sometimes times out in front of a prospect.

4.1 Demo plan by project

Project	What the demo must show	Recommended format	Gaps to close first
Guardian RAG	Synthetic policy pack (fictional company, ~10 documents, 3 domains); role switcher; question answered with citations; same question refused for another role; retrieval inspector showing which passages were searchable; audit log entry. Label the access-control panel clearly as a demonstration.	Hosted public demo (seeded accounts, nightly reset) + 3-min video + live local demo in meetings	Retrieval inspector view; demo data set; reset job; rate limiting
AutoFT	Monitoring dashboard; live drift chart crossing threshold after an injected topic shift; candidate vs incumbent table; trigger score breakdown; HOLD explained.	Streamlit app in <i>replay mode</i> (pre-computed results, no GPU) + video of the real training run	Fix the KeyError panel; replay data export; "simulated" label on any injected shift
KRONOS	Canvas with one end-to-end workflow: input → retrieval → LLM → output; live trace and execution log; run history replay.	Recorded walkthrough; live in meetings; source review on request	Pick and polish one workflow; script the demo
Swarminal	Real desktop: 3–4 panes, PTY behaviour (resize, scrollbar search), agent launch from preset, close-guard, browser preview.	2-minute screen recording + signed installer on GitHub Releases (<i>Mazen's decision</i>)	Licence file, product name, personal data out of screenshots
Nest Invest	Only with all contributors' consent: walkthrough of simulator and forecast service.	Case-study page once evidence in 2.6 exists	Everything in the 2.6 template

4.2 Demo deployment architecture RECOMMENDATION

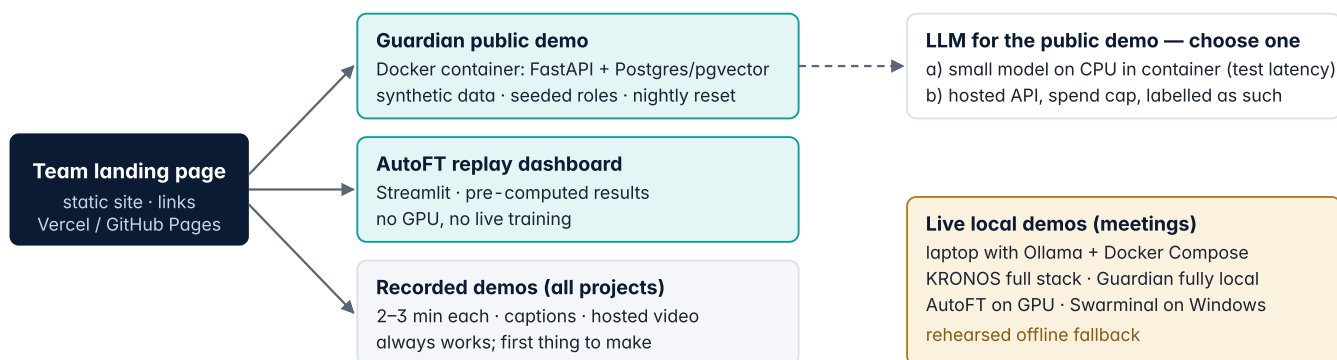


Figure 12. Public surfaces use only synthetic data and no GPU. Anything that needs a GPU, real local models or a desktop is shown live or on video.

4.3 Hosting options compared

Option	Cost and limits (official pages, 3 Oct 2026)	Best for	Watch out for
Streamlit Community Cloud	Free; shared resources up to 2 CPU cores and 2.7 GB RAM; apps sleep after 12 h without traffic ^[24]	AutoFT replay dashboard (already Streamlit)	Cold start after sleep; memory ceiling; public by default

Option	Cost and limits (official pages, 3 Oct 2026)	Best for	Watch out for
Hugging Face Spaces	Free CPU basic tier; paid CPU from \$0.03/h; GPUs from \$0.40/h; PRO account \$9/month with ZeroGPU quota [23]	Guardian demo as a Docker Space; ML audience visibility	Database inside the container is ephemeral — reseed on start
Vercel	Hobby free but for personal, non-commercial use; Pro \$20/month per user [26]	Landing page, Next.js front ends	A venture's marketing site is commercial — plan for Pro, or use a static host
Render	Hobby workspace \$0 with a limited free 512 MB web service; Pro workspace \$25/month; compute billed per second [25]	FastAPI backends with managed Postgres	Free tier limits; pgvector support should be checked for the plan used
Railway	30-day trial with \$5 credit; Hobby \$5/month minimum incl. \$5 usage; RAM \$0.00000386/GB-s, CPU \$0.00000772/vCPU-s [27]	Always-on small containers (Guardian API + Postgres)	Usage billing — set alerts
Local / recorded	No hosting cost	Swarminal; KRONOS; any GPU work	Rehearse; keep an offline copy of the video

Security considerations for public demos

- Synthetic documents only — never a client's or employer's material
- Seeded demo accounts; no self-registration; admin role hidden
- Rate limiting per IP; hard spend cap on any hosted LLM key
- Secrets in environment variables, never in the repo
- Nightly database reset; no persistence of visitor questions beyond aggregate counts
- Visible banner: "Demo with fictional data"
- Invite a prompt-injection challenge — and log results for the sales story
- Remove personal data from screenshots and videos

Implementation order

- 1 Week 1–2:** record Guardian and AutoFT videos from existing local builds; write synthetic policy pack.
- 2 Week 2–3:** add Guardian retrieval inspector; deploy Guardian Docker demo; deploy AutoFT replay on Streamlit.
- 3 Week 3–4:** landing page linking demos, videos and one-page briefs; KRONOS video.
- 4 Later, only if asked:** always-on paid hosting, GPU-backed live demos.

Demo readiness checklist

- ☐ Works from a fresh browser, logged out, on mobile
- ☐ Loads in under 10 s after a cold start (or warns)
- ☐ Every number on screen comes from a real run or is labelled "simulated"
- ☐ No errors visible anywhere on the page
- ☐ Fictional-data banner present
- ☐ Three scripted questions that always work
- ☐ One "refusal" question that demonstrates access control
- ☐ Video backup available offline
- ☐ Licence and attribution visible in the repo
- ☐ Owner's consent for any shared project
- ☐ Contact link on every demo page
- ☐ Someone outside the team has used it unassisted

PART 05

Branding, pitch and networking

Present the group honestly: a technical team with real systems, looking for the right problem — not a startup claiming traction it does not have.

5.1 Working names UNREGISTERED

None of these has been checked for trademark, company-name or domain availability. Run those checks in Egypt and in target export markets before using any of them publicly.

Name	Positioning	Intended audience
Proofline Labs	"AI you can verify" — cited answers, audit trails, promotion gates. Ties together the strongest trait across the portfolio.	Compliance-minded enterprises; technical buyers
Gatehouse AI	Governed access to company knowledge; a gate between documents and models.	HR, legal, compliance, IT security
Groundwork AI	Grounded, retrieval-first AI and the engineering groundwork beneath it.	Startups and SMEs buying AI engineering services
Nilegrid	Cairo-rooted systems engineering; neutral enough to cover services and products.	Regional clients; local ecosystem
Kemet Applied AI	Egyptian identity, applied rather than research focus.	Egyptian and Gulf clients; government programmes

5.2 Pitches

20-second introduction

"We're a small engineering team in Cairo that builds private AI assistants for organisations that can't send their documents to the cloud. Our assistant answers staff questions from company policy, cites the source, and never shows anyone a passage their role isn't allowed to see."

Technical capability statement

We design and deliver retrieval-augmented assistants with access control enforced inside retrieval and in the database; agent workflows with traces and evaluation gates; and LLM fine-tuning that ships only on held-out evidence. Everything runs on the client's infrastructure with local models by default. Front-end and API delivery in React, Next.js and Node; AI systems in Python and FastAPI.

60-second pitch

"Every company has policy documents staff need — leave, expenses, benefits — and some sections nobody outside HR or finance should see. Today the choice is a search box over everything, which leaks, or locked folders, which nobody reads. Cloud AI assistants make it faster but many regulated firms won't upload these documents.

We built Guardian RAG: it decides what a person may read *before* searching, enforces it again in the database, cites every answer, and logs every refusal for auditors. It runs fully on your own servers. In tests across six roles, restricted salary data was answered only for the roles cleared to see it.

We're not selling a finished product yet. We're looking for two or three organisations — ideally in healthcare, education or legal — to run a four-week pilot on their own policies. If it works, you keep it."

The test claim refers to the README's verified-behaviour table (3 documents, 29 passages) ^[3]. Do not generalise it beyond that scope when speaking.

Founder introduction — Yousef Mamdouh

Yousef is an AI/ML engineer in Cairo. Until mid-2026 he led multi-track AI R&D at YOXEL Technology — RAG security, autonomous fine-tuning and agent evaluation — and supervised a junior engineer. Before that he worked as a business data analyst on workflow automation and BI dashboards. His public projects are known for measuring before trusting: his fine-tuning system refused to ship its own model when the improvement fell inside measurement noise.

Projects in one sentence each

- **Guardian RAG** — policy answers with access control inside retrieval and a tamper-evident audit log.
- **AutoFT** — self-maintaining LLM agents: detects drift, retrains locally, and ships only if the new model measurably wins.
- **KRONOS** — a local-first platform for building, running and evaluating agent workflows.
- **Swarminal** (Mazen's) — a Windows desktop for running several CLI coding agents side by side.
- **Nest Invest** — graduation-project trading simulator with an LSTM forecasting service.

5.3 Networking strategy

Audience	Goal of the conversation	Opening question	Offer
Startup founders	Find AI features that are failing	"Which AI feature in your product do customers complain about most?"	Free 30-min retrieval/eval review
CTOs	Learn how they decide build vs buy	"How do you decide whether a model change actually helped?"	Evaluation audit
Engineering managers	Spot capacity gaps	"What's on your AI roadmap that nobody has time for?"	Scoped delivery
Corporate innovation	Identify a pilot sponsor	"Which documents would you never put into a cloud AI tool?"	Guardian pilot
Accelerator mentors	Pressure-test the segment	"If you were us, which industry would you talk to first, and why?"	Updates on progress
Investors	Learn, not raise	"What would you need to see from a team like ours in six months?"	Nothing yet
Recruiters	Keep employment options open	"Which applied-AI roles are hardest for you to fill?"	CVs and portfolio

TECHNE SUMMIT 2026 — VERIFIED SCHEDULE

The 2026 edition runs in two parts: **Cairo (GROW)** on 26–27 September at Ghurnata, Heliopolis — already past at the date of this report — and **Alexandria (START)** on 3–5 October at Bibliotheca Alexandrina, focused on early-stage startups across thirteen tracks, with the Startup World Cup regional competition on 4 October.

Organisers list 90+ workshops, one-to-one mentoring, AI-powered matchmaking and investor meetings^[18]; the agenda is on Sched^[19]. No contacts or meetings are assumed. If the team attends Alexandria, prioritise mentoring slots and accelerator representatives over investor pitches; if anyone attended Cairo, follow up those contacts within the week.

Follow-up message (within 48 hours)

"Thanks for the conversation at Techne about [their problem]. You mentioned [specific detail]. We've built a private policy assistant that enforces access control inside retrieval — here's a 3-minute demo: [link]. Would a 20-minute call next week be useful to see whether it fits [their context]? No pitch deck, just your documents' shape and our questions."

First meeting agenda (30 minutes)

1. Their context and current approach (10 min — they talk)
2. Discovery questions from 3.1 (10 min)
3. Short demo matched to what they said (5 min)
4. Agree a next step: documents to test, a contact in IT, or a polite no (5 min)

Write notes the same day in a shared interview log (Part 6).

5.4 Digital assets — what to build first

#	Asset	When	Notes
1	Guardian demo + video	Days 1–21	The single most persuasive asset
2	One-page project briefs (PDF)	Days 7–21	Guardian, AutoFT, KRONOS; consistent template
3	Team landing page + contact form	Days 14–28	One page: problem, demo, team, contact. No product claims
4	Professional email on own domain	Days 14–28	After a name is chosen and checked
5	Personal portfolios updated	Days 1–30	Each person keeps their own; link to team page
6	GitHub organisation	Only after IP agreement	Until then, keep repos under their owners; link, don't transfer
7	Individual product pages	After 3+ PoC requests	Not before there is demand
8	Pitch deck	After first pilot	A deck without traction invites the wrong conversation

PART 06

30 / 60 / 90-day execution roadmap

Roles below are proposals for discussion. They assume part-time availability that has not yet been confirmed by anyone.

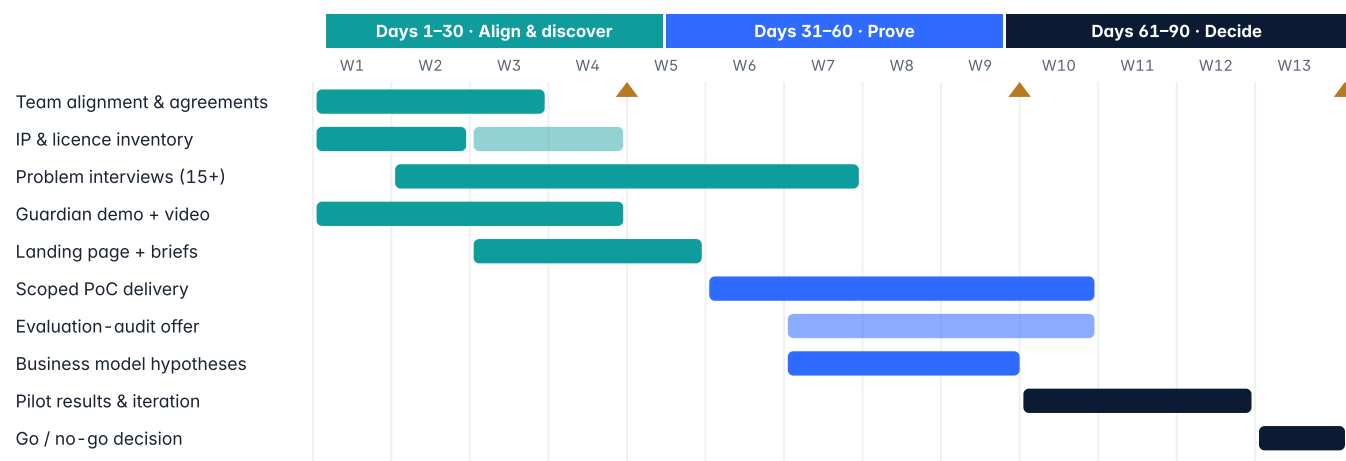


Figure 13. 13-week plan. Amber markers are decision gates at days 30, 60 and 90.

	Days 1-30 · Align & discover	Days 31-60 · Prove	Days 61-90 · Decide
Objectives	Agree how to work together; know what each person owns; one demo live; choose one problem	Run one narrow PoC; test a price; set a delivery process	Learn from pilot results; test repeatability; decide
Tasks	Alignment session; IP inventory (Part 7); synthetic policy pack; retrieval inspector; videos; 15 interviews; landing page	Scope PoC with one org; NDA; deploy on their infra; weekly demos; second-pass interviews; draft pricing; audit offer to 5 teams	Measure against acceptance criteria; client debrief; 2nd PoC or conversion to paid; decision memo
Proposed roles	Yousef: demo, interviews · Osama: inspector UI, landing page · Mazen: Swarminal release prep (own track) · Business advisor: interview targets	Yousef: PoC lead · Osama: client UI and integration · Advisor: pricing, contract	All: decision memo · Yousef: technical findings · Advisor: commercial findings
Deliverables	Signed collaboration terms; IP register; public demo; 3 briefs; interview log	PoC deployment; test results; pricing hypothesis; delivery checklist	Pilot report; case study (if permitted); go/no-go memo
Dependencies	Availability confirmed; employer-policy checks; legal consult booked	A willing organisation; their data and IT contact	PoC actually delivered
Success indicators	15 interviews logged; ≥ 3 PoC requests; demo used unassisted by 5 outsiders	Zero leaks in refusal set; client-judged accuracy met; client willing to pay or co-fund	Paid follow-on or referral; second org in pipeline
Decision gate	< 3 PoC requests → change segment or offer, not the code	No org will share data even under NDA → pivot to generic RAG services	Productise · continue consulting · apply to accelerator · pause

6.1 Weekly rhythm and lightweight project management

Cadence	Practice
Monday, 30 min	Plan: top 3 outcomes per person for the week
Async daily	One-line update in a shared channel: done / next / blocked

Realistic weekly load ASSUMPTION

Assuming roughly 8-12 hours per person per week alongside jobs and study:

- **Discovery:** 3-4 interviews per week, 45 min each plus notes
- **Build:** one demo-visible improvement per week

Cadence	Practice
Thursday, 45 min	Demo what changed; review interview notes
Every 2 weeks	Retrospective; update risk register
Days 30, 60, 90	Decision gate with written memo

Tools: one board (GitHub Projects or Jira — Yousef has Scrum/Jira experience), one shared drive for the interview log and IP register, one channel. Avoid more.

- **Admin:** 1 hour on legal, IP and finance tasks

At this capacity, one PoC at a time is the realistic limit. Taking two will slip both.

6.2 Decision framework — choosing the first offer

Score each candidate 1–5 on each criterion using *interview evidence*, multiply by the weight, and compare. The weights are a starting proposal; agree them as a team **before** scoring so the result is not reverse-engineered.

Criterion	Weight	Evidence that should drive the score
Pain intensity	25%	Interviewees describe a recent, costly incident unprompted
Willingness to pay or co-fund a pilot	25%	A named budget owner agrees to a priced next step
Speed to first value	15%	Pilot deliverable in ≤ 4 weeks with existing code
Team fit	15%	Skills evidenced in Figure 1; no single-person bottleneck
Repeatability	10%	Same problem heard from ≥ 3 organisations
Risk (inverse)	10%	IP clear; no regulated advice; data access realistic

RULE OF THUMB

If no candidate scores ≥ 3 on *both* pain and willingness to pay, the answer is "keep interviewing", not "pick the best of a weak set".

PART 07

Governance, ownership and risk

The questions below are cheap to settle now and very expensive to settle after a customer, accelerator or investor asks. This section identifies issues; it does not give legal advice.

7.1 Intellectual-property register (starting point)

Asset	Owner today	Licence	Open questions
Guardian RAG	Yousef	MIT ^[3]	Built independently of YOXEL work on "RAG security architecture"? Own time and equipment? Any employer code, designs or documents reused?
AutoFT	Yousef	MIT ^[4]	Same question for the "autonomous LLM fine-tuning system" track at YOXEL. Llama 3.2 Community License and dataset terms for commercial use.
KRONOS	Yousef	Apache-2.0 (stated) ^[5]	Private source under a permissive licence: if shared, anyone receiving it may reuse it. Decide whether a venture would need a different licence for new work.
Swarminal	Mazen	MIT claimed; file missing ^[7]	Add LICENSE; confirm the "BridgeMind-PTY" name is his own; any shared use needs a written agreement.
Nest Invest	Graduation team (incl. Yousef, Osama)	Unknown	All contributors and the university's policy must be identified before reuse.

HIGHEST-PRIORITY CHECK

Yousef's public profile describes YOXEL R&D tracks covering RAG security and autonomous fine-tuning (Oct 2025 – Jun 2026) ^[1]; Guardian RAG and AutoFT address the same themes. Overlapping *topics* are normal for an engineer's career and do not by themselves imply an employer claim. But before commercialising, Yousef should review his employment contract's IP-assignment and confidentiality clauses, document when and how each public repository was created, and obtain written confirmation from an Egyptian lawyer — and, if appropriate, a written release from the former employer. Osama, currently employed at YOXEL ^[10], should check outside-work and non-compete terms before contributing.

7.2 Three different kinds of agreement

Instrument	What it does	When to use it here
Project collaboration agreement	Sets out how people work together on a defined effort: contributions, who owns what is created, confidentiality, how income from a specific engagement is shared. No company is formed.	Now — covering demos, PoCs and any client work in the 90-day period.
Commercial licence	An owner keeps the IP and grants others defined rights to use it (scope, fees, duration, termination).	If Mazen agrees to Swarminal being used in a joint offer while remaining its owner; or if Yousef licenses Guardian to a client.
Company ownership structure	A legal entity owns the IP (by assignment), founders hold equity, usually with vesting; governance via shareholder agreement.	Only after the evidence gates in Part 9 are met.

7.3 Founder-readiness topics

Repository ownership	Keep repos under their owners until an entity exists; then transfer by written assignment. A shared GitHub org is not ownership.	Confidentiality	Mutual NDA among the team; standard NDA template for prospects.
Contributions to shared projects	Contributor terms (who owns contributions) before anyone commits to another person's repo.	Customer data	Process only on the client's infrastructure where possible; data-processing terms in every PoC; delete on completion with written confirmation.
Responsibilities	Write down each person's role, hours and deliverables per period.	Security	Secrets management; access reviews; the Guardian security test suite run on every client deployment; incident contact.
Equity discussions	Defer splits until roles are tested; base them on future commitment, not past projects alone.	Exit and handover	What happens to code, client relationships and shared accounts if someone leaves; documentation as a delivery requirement.
Vesting concepts	Equity earned over time, usually with an initial "cliff". Enforceability in Egypt to be confirmed.	Open-source obligations	Track licences of dependencies (e.g. AGPL tools are incompatible with some closed offerings); keep attribution files.
Decision-making	Who decides technical, commercial and spending questions; tie-break rule.		
Expenses and compensation	Who pays for hosting and tools; how engagement income is split; written record of every shared cost.		

QUESTIONS FOR EGYPTIAN LEGAL AND TAX COUNSEL

- Enforceability of IP-assignment and non-compete clauses in the founders' current or past employment contracts
- Choice of entity for a small founding team and its formation steps and costs
- Whether vesting and reverse-vesting arrangements are enforceable, or need an offshore holding structure
- Tax registration and invoicing obligations for service income received before an entity exists
- Obligations under Egypt's personal-data-protection framework (Law 151 of 2020 and implementing rules — confirm current status) when processing client documents
- University rules on ownership of graduation projects (Nest Invest)

7.4 Risk matrix

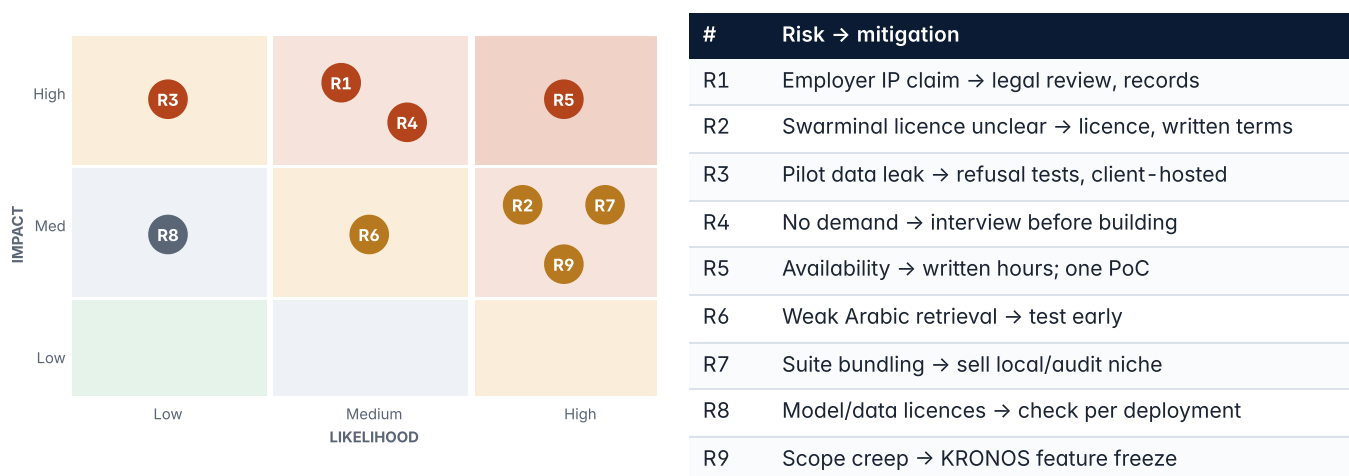


Figure 14. Placement is the authors' judgement from the evidence in this report.

PART 08

Financial and resource planning

No budget or revenue forecast is offered: there is no customer data to base one on. Instead, cost categories, three clearly labelled scenarios using official list prices, and the inputs a real model would need.

8.1 Cost categories

Category	What it covers	Price basis
Minimum viable infrastructure	Demo hosting, one small always-on container, a database	Official pricing pages — see scenarios [23] [24] [25] [26] [27]
Hosted LLM / embedding APIs	Only if the public demo uses a hosted model	Per-token; set a hard monthly spend cap and treat the cap as the budget line
GPU compute	AutoFT runs, client fine-tuning experiments	Hourly; HF Spaces GPUs from \$0.40/h [23] , existing RTX 5060 for internal work
Development tools	Repos, project board, CI	Free tiers are sufficient at this stage
Domain and email	One domain, 3–4 mailboxes	Registrar- and provider-dependent; not researched here
Specialists	Egyptian lawyer (IP, entity), accountant (tax registration), optional brand designer	Request 2–3 quotes each
Demo and pilot expenses	Travel to client sites, a test server or VM if the client lacks one	Per pilot; pass through to client where possible
Contingency	Unplanned costs	Planning convention of 15–20% on top of known costs

8.2 Three illustrative monthly scenarios ASSUMPTION-BASED

A · Zero-cash

\$0

+ domain and email

- AutoFT replay on Streamlit Community Cloud (free) [\[24\]](#)
- Guardian on HF Spaces free CPU tier [\[23\]](#)
- Recorded videos; static landing page
- Trade-off: sleep/cold starts; small model only

B · Lean always-on

≈ \$32–41

per month + domain, email, LLM cap

- Railway usage ≈ \$12.00 (below) [\[27\]](#)
- Vercel Pro \$20 for a commercial site [\[26\]](#)
- Optional HF PRO \$9 [\[23\]](#)
- Trade-off: small monthly bill; no GPU

C · Live GPU demo

\$8 – \$292

per month, GPU only

- 20 demo-hours × \$0.40/h = **\$8**
- Always on: 730 h × \$0.40/h = **\$292** [\[23\]](#)
- Conclusion: GPU on demand, never always-on, until a client pays

Scenario B — Railway usage arithmetic (30-day month = 2,592,000 s)	Rate [27]	Monthly
Guardian API container, 0.5 GB RAM, always on	\$0.00000386 / GB-s	0.5 × 2,592,000 × 0.00000386 = \$5.00
Postgres + pgvector container, 0.5 GB RAM, always on	\$0.00000386 / GB-s	0.5 × 2,592,000 × 0.00000386 = \$5.00
CPU, assumed 0.1 vCPU average across both (idle demo)	\$0.00000772 / vCPU-s	0.1 × 2,592,000 × 0.00000772 = \$2.00
Total usage (Hobby plan: \$5 minimum, includes \$5 usage)		\$12.00 billed

Egress (\$0.05/GB) ignored for a low-traffic demo. RAM and CPU averages are assumptions — measure real usage in week 1 and replace them. Prices in USD; convert at the rate on the payment date.

8.3 Bootstrapping and early revenue

Bootstrapping options ANALYSIS

- **Services first:** evaluation audits and scoped PoCs fund infrastructure and pay for legal set-up.
- **Client-funded pilots:** even a modest co-funded PoC is stronger evidence than a free one.
- **Pass-through infrastructure:** clients pay hosting and GPU directly.
- **Programme support:** accelerator and ecosystem programmes (e.g. ITIDA's Cairo Program ^[20]; ITIDA and Flat6Labs' Invest-IT ^[22]) once founders are committed. Terms must be read before applying.
- **Cloud credit programmes** offered by providers to startups — eligibility usually needs an entity.

Margin per engagement — formula, not forecast

margin = fee

- (person-days × internal day rate)
- pass-through costs not re-billed
- sales time (meetings, proposals)
- contingency

Track every engagement in this format from the first one. After three, the team will know its real delivery cost — the single most important number for any later financial model.

Information needed for a credible financial model

Input	How to get it in 90 days
Internal day rate per person	Agree in the alignment session; benchmark against 5–10 local quotes
Real hours available per week	Written commitment from each person
Conversion: interviews → PoC requests → paid	Interview log and pipeline board
Deal size	The first two quotes and their outcomes
Delivery cost per PoC	Time tracking on the first PoC
Infrastructure per client	Measured on the first deployment
Payment terms, taxes, currency	Accountant consultation; client's procurement rules
Support load after go-live	Tickets and hours in the first month after handover

BOTTOM LINE

The lean set-up costs between nothing and a few tens of dollars a month. The real constraint is **people's time**, not money — which is why Part 6 limits the team to one PoC at a time.

PART 09

Final recommendations

Answers to the seven questions in the brief. These are recommendations; none of them is a commitment by Yousef, Osama or Mazen.

Which project to demonstrate first?	Guardian RAG. Most complete public evidence, a risk anyone can understand in one sentence, a live role-switch demo that shows it, and the strongest fit with the team's verified skills (Figure 1, matrix 2.7).
Clearest service opportunity?	Permission-aware knowledge assistants (Guardian) for regulated mid-size organisations; second, LLM evaluation audits drawing on AutoFT's method.
Needs more research before commercialisation?	AutoFT (evaluation noise, no feedback data), KRONOS (unverifiable private source, no focused user), Swarminal (licence, naming, user validation — Mazen's decision), Nest Invest (all evidence missing).
Avoid building prematurely	Multi-tenant SaaS; a merged "platform of everything"; KRONOS features no client asked for; billing systems; mobile apps; a pitch deck before a pilot; a GitHub organisation before IP agreement.
Most important discovery questions	(1) When did someone last get a policy answer wrong, and what did it cost? (2) Which documents would you never put in a cloud AI tool? (3) Who would sponsor a 4-week pilot, and from which budget? (4) What would make it a success for you personally? (5) What have you already tried, and why did it stop?
Prepare before meeting a partner or investor	Live demo and video; one-page brief; IP register with legal confirmation; written team terms; interview log summary; pilot results or a clear statement that there are none yet; a specific ask.
Evidence that should trigger forming a startup	All of: (a) at least one paid or co-funded pilot that met its acceptance criteria; (b) two further organisations with the same problem asking for a pilot; (c) founders who have written down roles, hours and equity principles; (d) legal confirmation that the core IP can be assigned cleanly. Before that, a collaboration agreement is the right vehicle.

Prioritised action list

#	Action	Proposed owner	By	Status
1	Book an Egyptian lawyer: employment IP review (Yousef, Osama), entity questions	Yousef	Week 1	RECOMMENDED
2	Alignment session: hours, roles, decision rights, collaboration agreement draft	All three	Week 1	RECOMMENDED
3	Write synthetic policy pack; record Guardian and AutoFT videos	Yousef	Week 2	RECOMMENDED
4	Build Guardian retrieval inspector and demo polish	Osama (if available)	Week 3	RECOMMENDED
5	List 30 target organisations in healthcare, private universities and legal; start 15 interviews	Yousef + advisor	Weeks 2–6	RECOMMENDED
6	Fix AutoFT dashboard error; build replay mode	Yousef	Week 3	RECOMMENDED
7	Add LICENSE, Releases, consistent naming to Swarminal	Mazen (his choice)	Week 3	SUGGESTED
8	Landing page, one-page briefs, professional email	Osama + Yousef	Week 4	RECOMMENDED
9	Day-30 gate: pick the PoC target using the 6.2 framework	All	Day 30	RECOMMENDED
10	Recruit a business advisor or co-founder candidate	Yousef	By day 60	RECOMMENDED

EXECUTIVE ACTION BRIEF · OCTOBER 2026 · SHAREABLE

From engineering projects to a first customer

Prepared for Yousef Mamdouh. Summarises the full report; all claims there are labelled and sourced.

Where things stand

- Three of Yousef's projects have strong public evidence: **Guardian RAG** (access control inside retrieval, working), **AutoFT** (self-maintaining LLM agents: drift detection, local retraining, evidence-based promotion; experimental), **KRONOS** (broad agent platform, source private).
- **Swarminal** is Mazen Sarwat Ismail's project; any joint use needs his written agreement.
- **Nest Invest** lacks public evidence; treat as a credential.
- No customers, pilots or funding exist yet. The team is a **technical team**, not yet a startup.

The recommendation

Lead with **Guardian RAG as a self-hosted implementation service** for regulated mid-size organisations, starting with private healthcare groups and universities (hypothesis to test). Offer **LLM evaluation audits** as a second service. Freeze new KRONOS platform features for 90 days.

Next 90 days

Days 1–30	Legal IP check · team terms · public demo + videos · 15 interviews
Days 31–60	One scoped PoC on the client's infrastructure · priced offer · audit offer
Days 61–90	Pilot results · decide: productise, consult, accelerator, or pause

Gate for forming a company

A paid or co-funded pilot that met its criteria, two more organisations asking, written founder terms, and legal confirmation that the IP can be assigned.

Top risks

- Employer IP questions on themes overlapping prior R&D work
- No commercial lead in the team
- Crowded enterprise-RAG market — compete on local delivery and trust, not features
- Part-time availability

THIS WEEK

- 1) Book legal consult.
- 2) Hold team alignment session.
- 3) Start the synthetic policy pack and first demo video.
- 4) List 30 target organisations.

Contact: Yousef Mamdouh · github.com/YousefHlaly · Cairo, Egypt

APPENDIX

Project links, sources and access log

All sources accessed on **3 October 2026**. "Vendor" marks comparisons published by a company that sells a product in the category.

A.1 Supplied and project sources

1. Yousef Mamdouh — GitHub profile and README. <https://github.com/YousefHlaly>
2. Yousef Mamdouh — portfolio. <https://portfolio-yousefhlaly.vercel.app/>
3. guardian-rag repository (MIT). <https://github.com/YousefHlaly/guardian-rag>
4. AutoFT repository (MIT). <https://github.com/YousefHlaly/AutoFT>
5. kronos-showcase repository (Apache-2.0; case study, source private). <https://github.com/YousefHlaly/kronos-showcase>
6. Swarminal repository — Mazen Sarwat Ismail. <https://github.com/sspotter/Swarminal>
7. Swarminal LICENSE link from README — returned HTTP 404. <https://github.com/sspotter/Swarminal/blob/main/LICENSE>
8. Mazen Sarwat Ismail — GitHub profile. <https://github.com/sspotter>
9. Osama Ashur — GitHub profile. <https://github.com/Osama-Ashur>
10. Osama Ashur — portfolio. <https://osama-ashur-portfolio.vercel.app/>
11. Osama Ashur — LinkedIn. Not retrievable (robots.txt); not used. <https://www.linkedin.com/in/osama-ashur/>

A.2 Market, ecosystem and pricing sources

12. Onyx — "Enterprise RAG platforms 2026" (vendor, May 2026). <https://onyx.app/insights/enterprise-rag-platforms-2026>
13. AgentsRoom — "Best tools to run multiple coding agents in 2026" (vendor, June 2026). <https://agentsroom.dev/blog/best-multi-agent-coding-tools>
14. Conductor — product site. <https://www.conductor.build/>
15. FutureAGI — "Dify vs Flowise vs Langflow 2026" (vendor). <https://futureagi.com/blog/dify-vs-flowise-vs-langflow-2026/>
16. FlowiseAI/Flowise repository — archived 13 Aug 2026. <https://github.com/FlowiseAI/Flowise>
17. FutureAGI — "Best AI drift detection tools 2026" (vendor). <https://futureagi.com/blog/best-ai-drift-detection-tools-2026/>
18. Egypt Innovate — "Techne Summit 2026 returns across Cairo and Alexandria". <https://egyptinnovate.com/en/news/techne-summit-2026-returns-across-cairo-and-alexandria-with-more-than-us1-million>
19. Techne Summit 2026 agenda (Sched). <https://technesummit2026.sched.com/>
20. Egypt Innovate — ITIDA Cairo Program, 8th cohort for AI startups (July 2026). <https://egyptinnovate.com/en/news/itida-opens-applications-for-the-8th-cohort-of-the-cairo-program-to-support-egyptian-ai-startups>
21. Flat6Labs — programmes. <https://www.flat6labs.com/programms/>
22. ITIDA — Invest-IT programme press release (listed in search; not reviewed in detail). <https://itida.gov.eg/English/PressReleases/Pages/itida-launches-invest-it-startup-funding-program-egypt.aspx>
23. Hugging Face — pricing. <https://huggingface.co/pricing>
24. Streamlit — Community Cloud resource limits. <https://docs.streamlit.io/deploy/streamlit-community-cloud/manage-your-app>
25. Render — pricing. <https://render.com/pricing>
26. Vercel — pricing. <https://vercel.com/pricing>
27. Railway — pricing. <https://railway.com/pricing>

A.3 Image credits

Figures 3, 5–9 are screenshots published in the respective public repositories by their owners (Yousef Mamdouh; Mazen Sarwat Ismail), reproduced for review and commentary with attribution; cropping and blurring of personal identifiers were applied for this report. Figures 1, 2, 4 and 10–14 are original diagrams created for this report from the sources above. No stock imagery is used.

A.4 Quality-control log

Check (from the brief)	Result
Supplied links verified	All opened; LinkedIn blocked; Swarminal LICENSE 404; Nest Invest has no public source
Project claims vs sources	Every project figure traced to its README or portfolio; none reproduced by execution
Facts vs hypotheses separated	Labels applied throughout
Diagrams vs implementation	Architecture figures redrawn from README diagrams; integrations shown as possibilities only
Calculations; rendered PDF	Part 8 arithmetic shown in full; every page inspected before delivery; Swarminal attributed to Mazen throughout